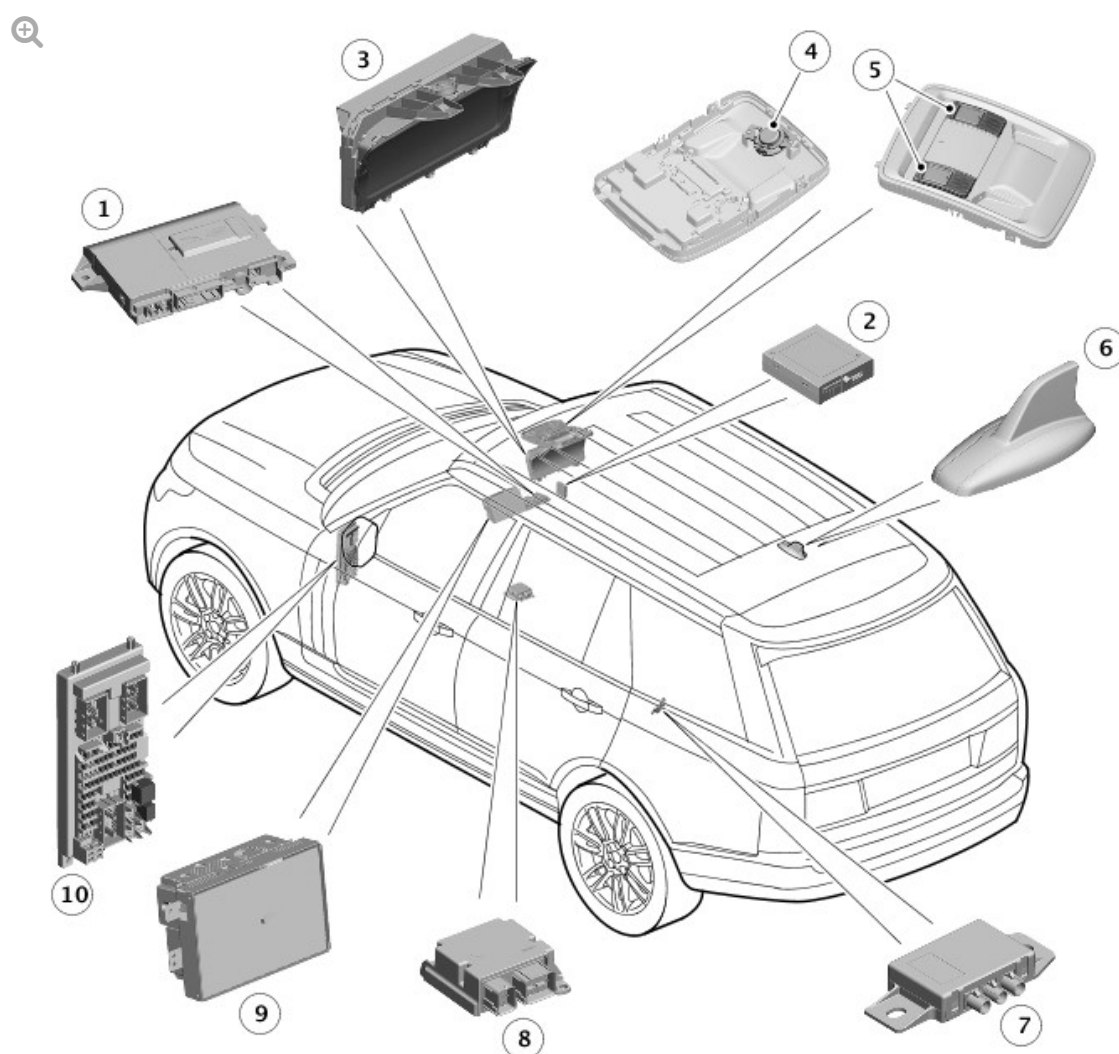


2016.0 RANGE ROVER (LG), 415-01

INFORMATION AND ENTERTAINMENT SYSTEM

DESCRIPTION AND OPERATION

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Telematics Control Module
2	SIM Card Module
3	Instrument Cluster (IC)
4	Telematics speaker
5	Emergency (eCall)/Breakdown (bCall) switches - overhead console
6	Roof pod - GSM/GPS antenna
7	GPS signal splitter
8	Restraints Control Module (RC)
9	Touch Screen (TS)
10	Central Junction Box (CJB)

OVERVIEW

Telematics is a system of devices and features which communicates with other devices or with human users over several communications networks including vehicle networks and cellular networks Global System for Mobile communications (GSM).

The telematics system is integrated into the vehicle powertrain and body MultiCAN systems and communicates with the vehicle infotainment system. The system is controlled by a Telematics Control Module.

Land Rover's telematics system is marketed under the name of 'InControl™'

Telematics is a combination of built-in mobile connectivity, global satellite location information and vehicle status and control. The system provides:

- Emergency Call Support
- Breakdown Call
- Stolen Vehicle Tracking/Recovery

- Mobile phone applications.

The telematics system also features an in-vehicle WiFi hotspot using access to a 3G cellular network. The system enables the customer to access an internet connected WiFi service via the Telematics Control module and a SIM Card Module which contains the users SIM card.

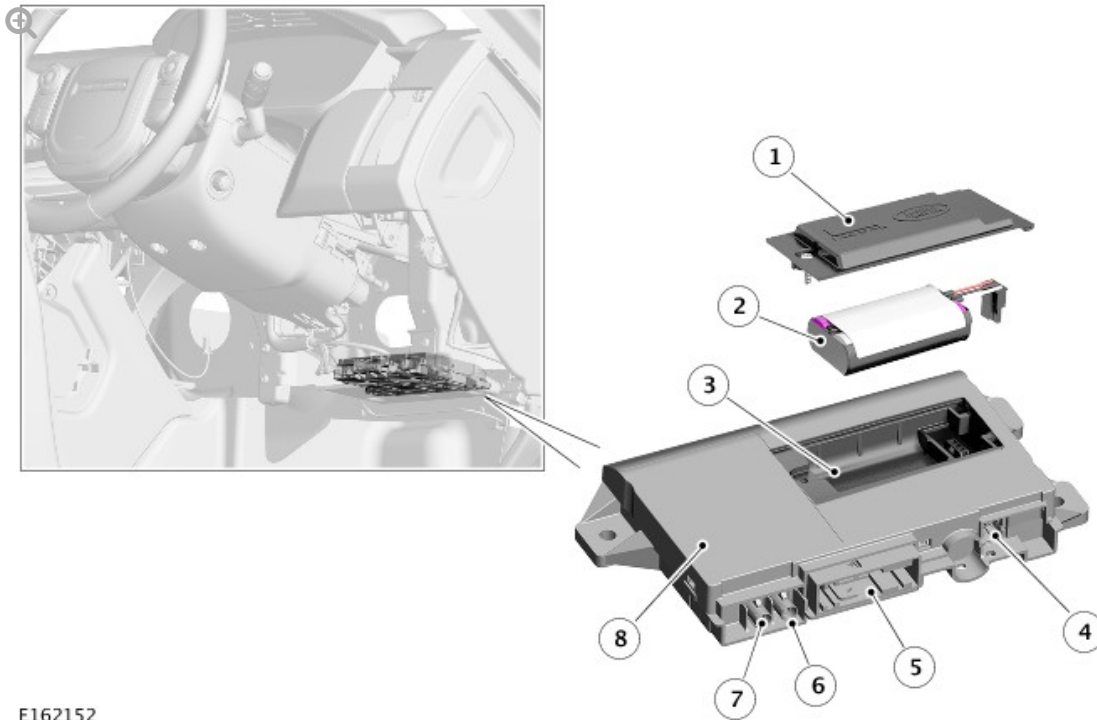
For NAS customers, the SIM card is provided to the customer and they cannot register to their network of choice.

The WiFi hotspot provides the following features:

- In-vehicle internet connection for WiFi enabled devices
- A Local Area Network (LAN) for more than one device, for example; for gaming
- The internet connection is provided by the Telematics Control Module (TCU), using a customer provided Subscriber Identification Module (SIM) card located in the SIM Card Module.

DESCRIPTION

TELEMATICS CONTROL MODULE



E162152

ITEM	DESCRIPTION
1	Battery cover
2	Battery pack
3	Battery tray
4	SIM Card Module connector
5	Electrical connector
6	GPS antenna connector
7	GSM antenna connector
8	Telematics Control Module (TCU)

The TCU is located below the steering column, behind the instrument panel lower trim panel. The TCU is attached to a bracket which in turn is attached the instrument panel cross-car beam.

For security reasons the TCU is located in a position that is not easy to access. This deters the action of a thief to remove or disconnect the module. Ease of disconnection would disable the tracking facility.

The TCU controls the Telematics system, communicating with the vehicle systems over High Speed (HS) Controller Area Network (CAN) powertrain systems bus and the Medium Speed (MS) CAN body systems bus. The system transmits/receives signals via RF (Radio Frequency), Global Positioning System (GPS) and GSM.

The TCU has an integral embedded Subscriber Identification Module (SIM) card. The embedded SIM card is part of the module circuitry and cannot be removed. The embedded SIM card is configured to enable the network to retrieve and pass information to the Telematics Service Provider (TSP), this enables the TSP to call back the TCU. During an eCall and bCall, the embedded SIM card will be used, overriding the SIM card in the SIM Card Module. This will result in loss of WiFi during the call and call back.

A back-up battery pack, consisting of two battery cells, is integrated into the TCU. The battery pack enables continued operation in the event of a vehicle main power source dis-connection; for example in a major accident or in the case of deliberate action of a thief as part of a vehicle theft attempt. This feature enables the eCall and vehicle tracking functions to carry on operating in the above mentioned circumstances.

Telematics Control Module Battery Pack



E160290

The battery pack is located in the TCU, under a cover which is retained by a screw.

The battery pack contains one re-chargeable cell and one non-rechargeable cell. The two battery cells are controlled separately by the TCU software.

NOTE:

The two battery cells cannot be replaced as separate items. The battery pack is replaced as a complete unit.

The use of two different battery cells provides continued performance at very low temperatures. The re-chargeable cell cannot deliver the charge effectively at very low temperatures. The non-rechargeable cell has improved performance at low temperatures.

The back-up battery will only be active during an eCall, bCall or stolen vehicle tracking session in which the vehicle power supply is disabled. Once the non-rechargeable cell has been used it will require replacement to restore the full capability of the eCall and stolen vehicle tracking systems.

TCU battery back-up power is also required for the Land Rover logistics tracking system when the vehicle is in transit mode.

Transit mode removes power to the TCU approximately 15 seconds after the ignition is switched off (power mode 0). The ignition off mode may occur before the TCU has sent all of its logistics tracking data to the manufacturer servers.

The backup battery enables the TCU to reliably deliver this data. The TCU is programmed not use the non-rechargeable cell for this purpose.

The TCU is in a pre-activated state when in transit mode. The non-rechargeable battery function is not activated as it would become depleted during transportation to the dealer and would immediately require a service replacement before handing the vehicle over to the customer. For this reason only the re-chargeable cell is used in these circumstances.

During the vehicle Pre-Delivery Inspection (PDI) process the TCU is removed from transit mode and switched on to an 'activated state' where both battery cells become active. During a TCU main power source disconnection the re-chargeable battery cell will always be used first. The non-rechargeable cell is only used if eCall or stolen vehicle tracking are activated.

The expected battery pack life is approximately 10 years if not used frequently.

The TCU charges the re-chargeable cell when the ignition is on (power mode 6). The charging function is controlled and regulated by the TCU software.

If either battery cell becomes depleted below a preset threshold or fails completely the following warnings/actions are displayed in the message center:

- SOS System Limited
- SOS System Not Available

A signal is sent to the Telematics Service Provider and a Diagnostic Trouble Code (DTC) is set in the TCU

When the back-up battery has been replaced, the system must be re-set using a specific application on a Land Rover approved diagnostic system, setting the battery level back to 100%.

SERVICE MODE

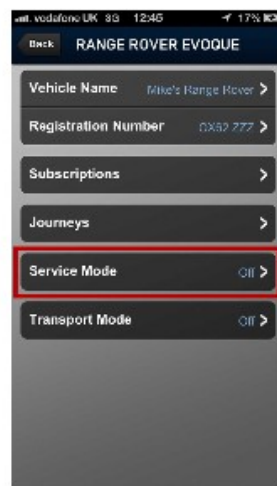
Before disconnecting power to the TCU, for example disconnecting the primary vehicle battery to carry out maintenance repairs, the vehicle owner must place the TCU into '**Service Mode**'.

If the disconnection procedure is not carried out, the system will activate the InControl Secure application and will enter 'Theft Notification' or 'Stolen Vehicle Tracking' mode during which time the charge in the back-up battery

would be used. If the non-rechargeable back-up cell becomes depleted through this action, it may reach the point where it requires replacing.

For security purposes 'Service Mode' is the sole responsibility of the vehicle owner.

During vehicle maintenance, where the primary vehicle battery is disconnected for a period of time exceeding the 10 hour window, the vehicle owner must either be informed by the dealer to re-activate Service Mode (using their mobile phone app) or the owner must inform the 'Stolen Vehicle Tracking' service provider that the vehicle is currently in the dealership for repairs with the vehicle battery disconnected, instructing the service provider to ignore the Stolen Vehicle Tracking notifications during the repair period.



E160473

TRANSPORT MODE

The TCU contains a motion sensor which is used for vehicle movement detection as part of the stolen vehicle tracking system. Prior to transporting the vehicle, for example a ferry journey, the vehicle owner must place the TCU into '**Transport Mode**' to disable the motion sensor.

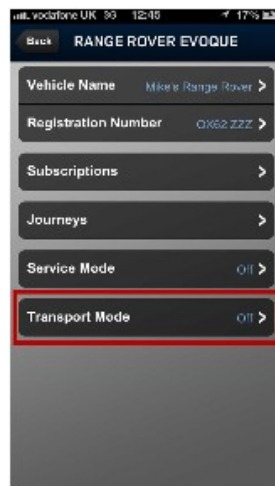
If the motion sensor is not disabled and the ignition is switched off (power mode = 0), but vehicle movement is detected, the 'Theft Notification' function will be activated. As with Service Mode, Transport Mode is the sole responsibility of the vehicle owner for security reasons.

The motion sensor is capable of detecting slow movement, e.g. jacking the vehicle to remove a road wheel, or winching onto a transporter.

Triggering of the sensor does not immediately generate a 'Theft Notification', but it will activate the GPS receiver and measure the vehicle's movement. If the vehicle moves more than a predefined distance (approximately 200 meters) the TCU sends a theft alert to the service provider.

The service provider contacts the vehicle owner to determine if a genuine theft has taken place. If a vehicle theft is confirmed the service provider initiates full GPS tracking of the vehicle.

Transport Mode disables the movement assessment by the GPS that follows a motion detection, therefore no 'Theft Notification' will be sent to the service provider. Should transporting the vehicle exceed the 10 hour window it is the responsibility of the vehicle owner to reset Transport Mode to avoid a false theft notification being generated.



E160474

When the vehicle has the ignition off (power mode 0) and the MultiCAN buses are 'asleep', the TCU is still active, enabling movement detection should an illegal attempt be made to move the vehicle. The TCU has a permanent battery feed to enable this function. The drain on the vehicle primary battery is minimal and the vehicle remains well within its quiescent drain target.

The TCU is linked to the engine immobilization circuit. When a start request is made the Central Junction Box (CJB) requests the TCU to pass a secure identification over the MultiCAN bus. If the identification matches the one in the CJB, the start request is authorised.

In the event of disconnection or damage to the roof pod or the Telematics antenna, resulting in the loss of the GSM signal, an internal antenna, integrated into the TCU, is used as a back-up GSM antenna. The backup GSM antenna is used to re-establish communication with the emergency service provider or TSP.

SIM CARD MODULE



ITEM	DESCRIPTION
1	USB port
2	SIM card holder release switch
3	SIM card holder
4	SIM card

The SIM Card Module is located below the steering column, behind the instrument panel lower trim panel, adjacent to the TCU. The SIM Card Module is attached to a bracket which in turn is attached the instrument panel cross-car beam.

The SIM Card Module is used to establish the InControl Wi-Fi feature. The SIM Card Module contains the users SIM card which enables the system to register to a network of personal choice.

For NAS customers, the SIM card is provided to the customer and they cannot register to their network of choice.

The SIM card is placed in the SIM Card Module which is directly connected to the TCU. The SIM card reader utilises a 'Mini-SIM' interface. If the SIM card is a different size, e.g. 'Micro-SIM' an adaptor or replacement SIM card will be required.

To install a SIM card, press the SIM card holder release switch to open the SIM card holder. Insert the SIM card into the holder and close the holder.

NOTE:

If the SIM card has been previously used in a mobile phone or other device and a Personal Identification Number (PIN) has been set, the PIN must be removed before use in the vehicle.

The USB connector in the SIM Card Module is connected to the TCU for transfer of data between the modules.

FRONT OVERHEAD CONSOLE



E162153

ITEM		DESCRIPTION
1		Telematics speaker
2		Breakdown call (bCall) switch
3		Emergency call (eCall) switch
4		Microphone

The front overhead console houses the Breakdown call (bCall) and Emergency call (eCall) switches, microphone and Telematics speaker. The functions of the bCall and eCall switches are controlled by the TCU.

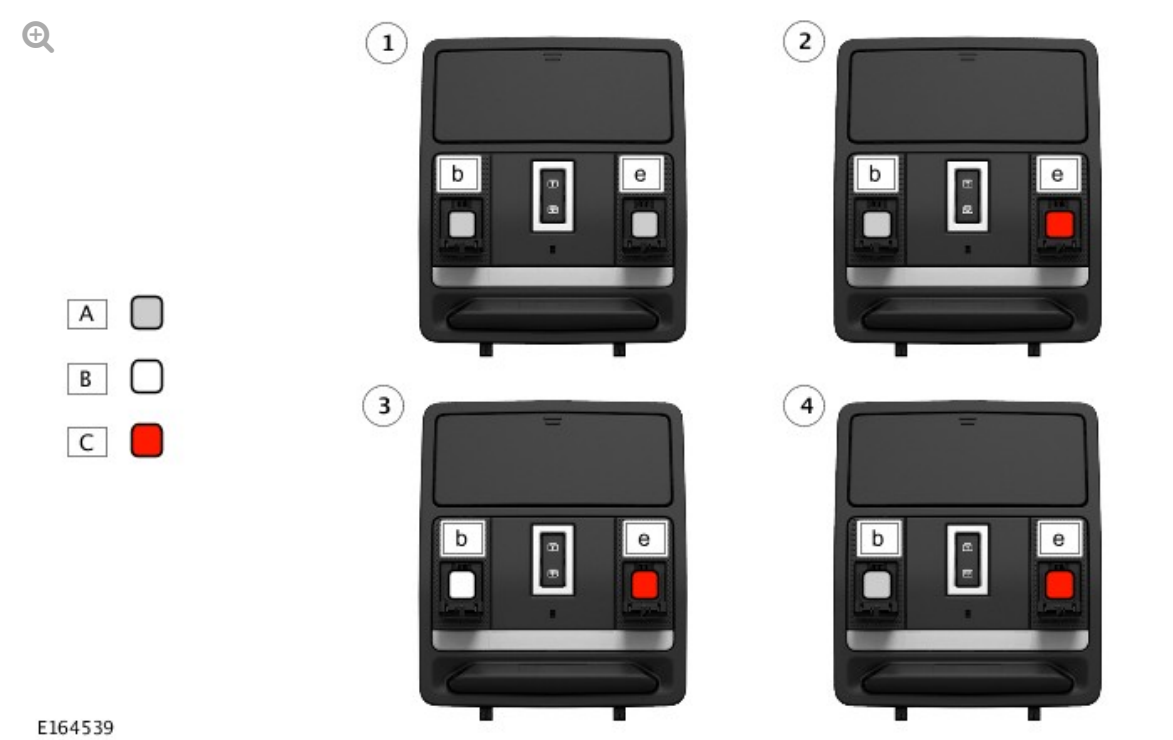
A diagnostic line (BTN), between the front overhead console and the TCU, enables the system to check the harness wiring is connected and the interface is functioning correctly. In operation, the TCU briefly pulls the diagnostic line (BTN) to ground, allowing the TCU to detect the circuit completion.

The microphone input is shared by the TCU and Integrated Audio Module (IAM). The TCU controls the output to the Telematics speaker.

LED lamps, integrated into the eCall and bCall switches, provide switch illumination and indicate the status of the call operation.

The following illustration shows the LED lamp status for the eCall and bCall switches during normal use.

Emergency Call and Breakdown Call Switch LED Lamp Status

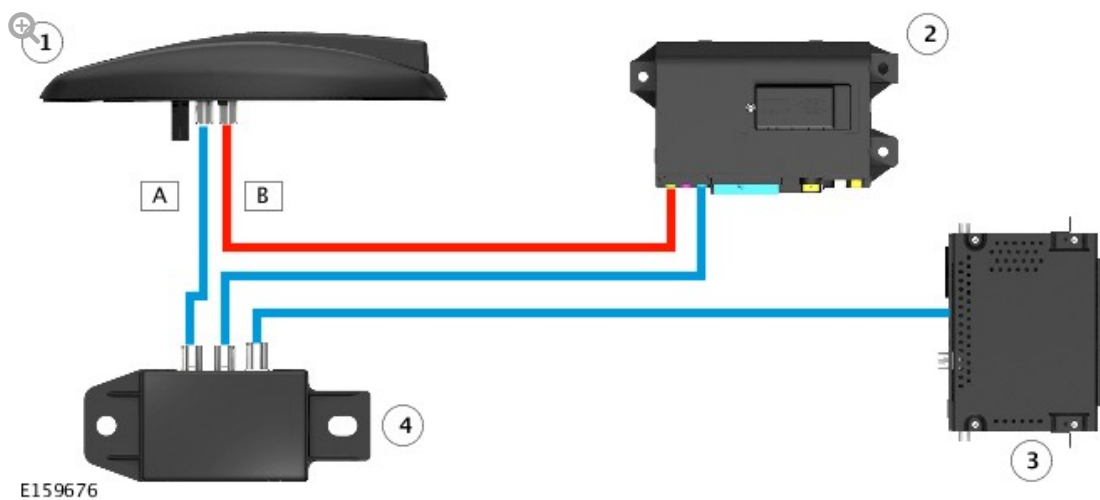


ITEM	DESCRIPTION
A	LED lamp extinguished
B	bCall - permament white lamp
C	eCall - permanent red lamp
b	b-call switch
e	e-call switch
3	State 1
4	State 2
b	State 3
e	State 4

Prior to activation the current status if InControl Remote, Secure and Wifi can be initially identified from the Light Emitting Diode (LED) lamp sequences of the e-Call and b-call switch lamps in the front overhead console.

- The lamp LED lamp sequences denote the activation status:
- **State 1** - The CAN bus is awake or the engine is running but neither of the LED lamps are illuminated
 - The vehicle has not gone through a Pre-Delivery Inspection (PDI). InControl Remote, Secure and Wi-Fi cannot be activated until the PDI is complete.
- **State 2**
 - The CAN bus is awake or the engine is running. The e-Call switch is illuminated (red). The b-Call switch is not illuminated.
 - The vehicle PDI is completed, InControl Remote, Secure and Wi-Fi is ready for activation.
- **State 3**
 - The CAN bus is awake or the engine is running. The e-Call switch is illuminated (red), the b-Call switch is illuminated (white).
 - InControl Remote, Secure and Wi-Fi activation completed and is linked to a user account.
- **State 4**
 - The CAN bus is awake or the engine is running. The e-Call switch is illuminated (red). The b-Call switch is not illuminated (post activation).
 - The user account has been de-registered from the vehicle. The account can now be connected to a different user.

GLOBAL POSITIONING SYSTEM (GPS) SIGNAL SPLITTER



ITEM	DESCRIPTION
A	GPS signal
B	GSM signal
1	Roof pod (GPS/GSM antenna)
2	Telematics Control Module
3	Integrated Audio Module (IAM)
4	GPS signal splitter

ANTENNA

The roof pod on vehicles with the Telematics system contains antennas for the reception of GPS signals and Global System for Mobile communications (GSM) signals. Depending on vehicle specifications, the roof pod can also contain antennas for SDARS and Digital Audio Broadcast (DAB).

For additional information, refer to: [Antenna](#) (415-01 Information and Entertainment System, Description and Operation).

The GPS signal is passed via a co-axial cable to the GPS signal splitter where it is then passed to the Integrated Audio Module (IAM) for use by the navigation system and to the TCU for use by the Telematics system.

OPERATION

The TCU receives the Global System for Mobile communications (GSM) and Global Positioning System (GPS) signals via the roof pod.

The GPS signal is received by the roof pod antenna and is passed to the GPS signal splitter. The GPS output from the GPS signal splitter is sent to the TCU for Telematics system use and to the Integrated Audio Module (IAM) for navigation use.

The GSM signal is received by the roof pod antenna. The signal is passed by a co-axial cable directly to the TCU. In the event of a malfunction of the roof pod, the backup GSM antenna within the TCU will also receive the GSM signal.

The GSM signal provided by Vodafone enables the TCU to communicate with the Telematics Service Provider (TSP) Wireless Car. The TSP provides secure mobile data connection and co-ordinates the sending and receiving of vehicle data and control commands. Telematics provides two main services; vehicle and driver.

The Land Rover 'InControl' system includes vehicle services; Breakdown call (bCall), Emergency call (eCall) and stolen vehicle tracking. Driver services include mobile phone applications with remote vehicle functions.

Several third party suppliers are integral to the InControl system. In each market emergency response agencies are integrated with InControl along with Land Rover Assistance and Stolen Vehicle Tracking.

A dedicated mobile phone application and the InControl owner website are available to provide an intuitive and secure interface for InControl users. An administration website is available for dealerships to set up and manage customer accounts.

The TCU receives the following inputs:

- Hardwired and CAN crash signal from the Restraints Control Module (RCM) - automatic e-Call activation.

- eCall/bCall switch activation signals
- Microphone signal
- Roof pod antenna GSM and GPS signals.

The TCU provides the following outputs:

- Telematics speaker.

Communication with the vehicle systems is via the MultiCAN buses, enabling remote vehicle feature functions. The following TCU primary communications transmit data on the following MultiCAN buses:

- Engine immobilisation start request ID - TCU to Central Junction Box - HHS CAN powertrain systems bus
- RCM crash signal status/verification - TCU to RCM - HS CAN powertrain systems bus
- Land Rover InControl data - TCU - MS CAN body systems bus via GateWay Module (GWM) - MS CAN body systems bus to Instrument Cluster (IC) and Touch Screen (TS).

RESTRICTED WIFI DURING ECALL/BCALL

During an eCall and bCall, the embedded SIM card in the TCU will be used, overriding the SIM card in the SIM Card Module. This will result in loss of WiFi during the call and call back.

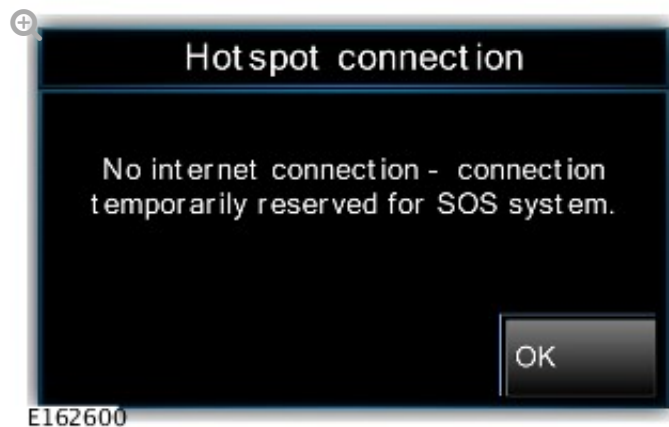
The embedded SIM card is used for the following reasons:

- The TCU embedded SIM card subscription will never expire. The customer SIM card may have restrictions or lack of credit which would make a bCall impossible. Note that eCall is an emergency call and for this reason it is a free to use service.
- The embedded SIM card in the TCU is a way that the network can retrieve and pass information to the Telematics Service Provider (TSP). This enables the TSP to call back using the embedded SIM card in the TCU on a known number. Some customer SIM cards may not disclose the relevant

information to allow the call back.

- During the call-back period, to allow the TCU to be able to pick up the call which is addressed to the TCU embedded SIM card phone number, the TCU will remain registered to the TSP network using the embedded SIM card.

During an eCall or bCall or the during the call back period, the following image will be displayed in the TS.



TRANSPORT AND SERVICE MODES

When the vehicle is being serviced or repaired, InControl Secure must be set to 'Service Mode'. Service Mode temporarily disables the Stolen Vehicle Tracking function which becomes active when the vehicle primary battery is disconnected.

If the vehicle is being transported, InControl Secure must be set to 'Transport Mode'. Transport Mode temporarily disables a motion sensor in the TCU and therefore the Stolen Vehicle Tracking function is temporarily disabled.

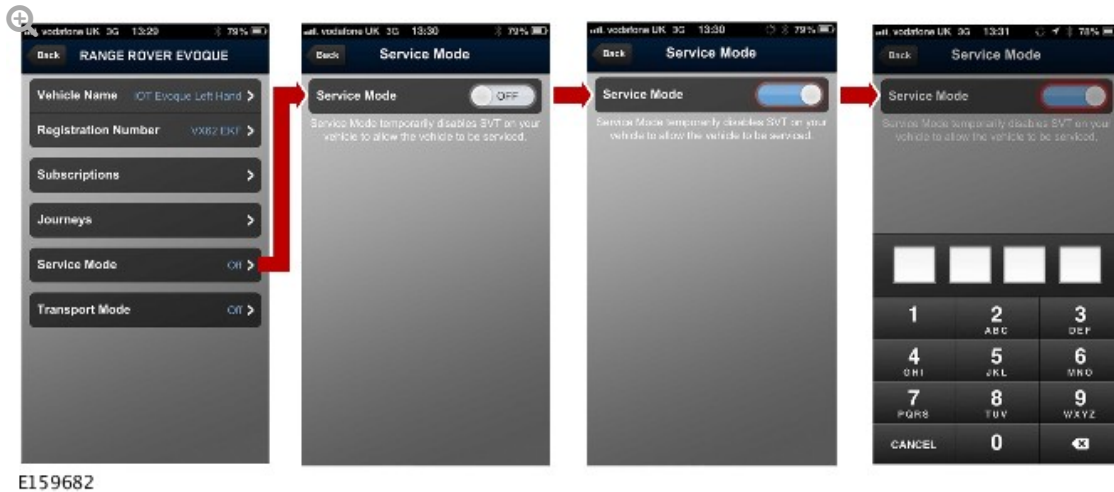
Service and Transport Mode are activated using the InControl Secure remote mobile phone application, or via the owner website.

An example of Service Mode activation is shown in the following illustration. The user must enter their Land Rover InControl account PIN as requested to complete the activation request. Once activated, the screen slider switch

changes to red, indicating Service Mode is engaging. Transport Mode is activated in the same manner.

NOTE:

The Service and Transport Modes are set for 10 hours. If this time period is exceeded the modes must be set again if required.



INCONTROL WI-FI FEATURES

This option of the Telematics system allows the user in-vehicle access to 3G cellular networks. Up to 8 devices are supported. More than 8 devices can be connected but the speed of the network compromised.

The cellular internet connection is provided by the TCU using a user provided SIM card. The WiFi access and data plan is tied to a mobile internet provider or specific carrier's network.

The SIM card is placed in the SIM Card Module which is directly connected to the TCU.

Access to activate the WiFi features is via the Extra Features and WiFi Hotspot settings screens on the vehicle TS.



A



B



E160414

ITEM	DESCRIPTION
A	Extra features menu screen
B	WiFi Hotspot menu screen

The following basic and advanced settings are accessible from the WiFi menu screen:

- Hotspot name - Cannot be changed. It is fixed and derived from the vehicle model plus the last 6 digits of the Vehicle Identification Number (VIN). This provides a unique network ID for each vehicle.
- Password - Can be changed in WiFi advanced settings menu to a user preference.
- Hotspot and data roaming - On/Off soft keys
- Help soft key - Access to advanced WiFi settings
- Auto generate soft key - Automatically generates a new random password.



ITEM	DESCRIPTION
A	Hotspot name
B	Password
C	Hotspot and data roaming soft keys
D	Help soft key
E	Auto generate soft keys

The following information and options are available in the 'Help' menu:

- Hot spot router address
- Restore default router settings
- Information regarding changing WiFi settings.

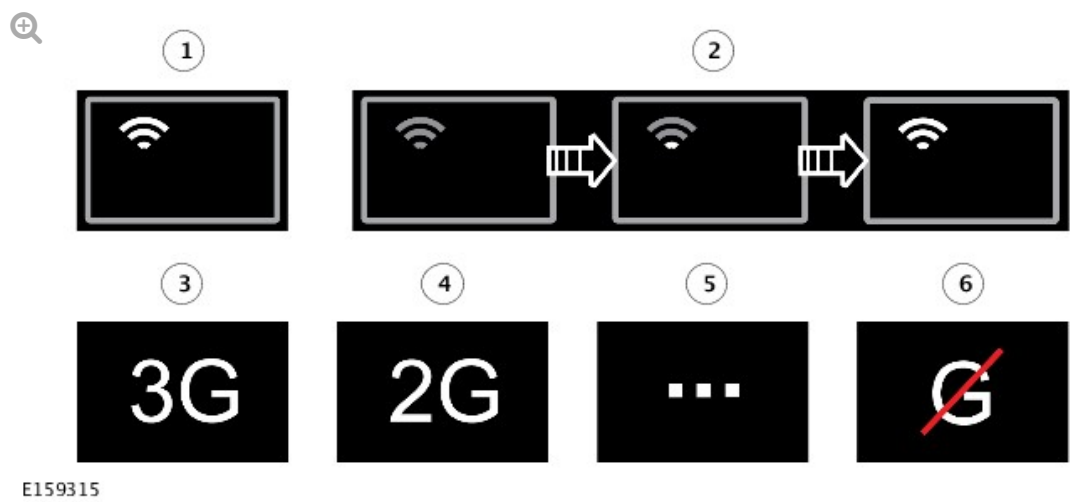
Advanced settings, including changing your security password and Access Point Name (APN) configuration, are performed by logging into the hotspot router address (using a device such as a mobile phone or laptop). This allows access to the administration network portal to make the required changes. If the user has forgotten the password and hotspot name, the 'Restore Default' soft key will reset the system with a random password and hotspot name to allow continued operation of the system. The password can also be reset by using the 'Auto Generate' soft key. The 'Reset' soft key will not reset the hotspot name, this is fixed. It will however, reset all the

other settings accessed through the portal on the customers device.



ITEM	DESCRIPTION
A	Hot spot router address
B	Router settings - Restore default - soft key

The WiFi icons, shown in the following illustration, indicate the status of the WiFi hotspot signal. The icons are viewed in the top right hand side of the TS.



ITEM	DESCRIPTION
1	WiFi on

2	WiFi initialising
3	3G mobile network connectivity
4	2G mobile network connectivity
5	Connecting
6	No mobile network connection

FAULT DETECTION

The following is a list of possible faults:

- If the Telematics Control Module fails to configure, contact Dealer Technical Support (DTS) to confirm the server is operational before attempting further diagnosis or repair.
- 'No SIM inserted' message - allow TCU at least 1 minute to read the SIM card. If error persists connect Land Rover approved diagnostic equipment to the vehicle diagnostic connector. A DTC will be set for 'no SIM Card Module connected'.
- WiFi not available due to permanent eCall/bCall - connect diagnostic equipment to check for SOS On Call Keypad DTC
- Date incorrect on attached devices list - connect diagnostic equipment to check for GPS Antenna DTC
- WiFi Selected, no hotspot available - connect diagnostic equipment to check for Antenna DTC
- TS setting conflict, for example WiFi off, TS icon shows WiFi on - connect diagnostic equipment and check for MS CAN body systems bus DTC.
- No network coverage - check vehicle is in area where good reception would be expected. Check that SIM card works in a known device, for example; customers mobile phone.
- SIM card:
 - Check that SIM card has been correctly inserted into SIM card holder in SIM Card Module.

- Check that SIM card has credit with network provider.
- Has the SIM card data allowance been exceeded?
- If abroad, has data roaming been authorized by the network provider?
- Has SIM card been activated with the network provider?
- Has a PIN been set on the SIM card? Message will be displayed on TS. Owner must insert SIM card in mobile phone and disable the PIN.
- WiFi Network Connection:
 - What device is being used to connect to the network? 802.11b/g/n WiFi types supported. WiFi Hotspot operates in the 2.4GHz band only. 5GHz band, 802.11a or 802.11ac devices are not supported.
 - What year was the device made or purchased? The security protocol for password type is set to WPA2, if the device doesn't support WPA2 then the customer will have to change this via the local host page.
NOTE: All WiFi devices from 2006 should support WPA2 protocol.

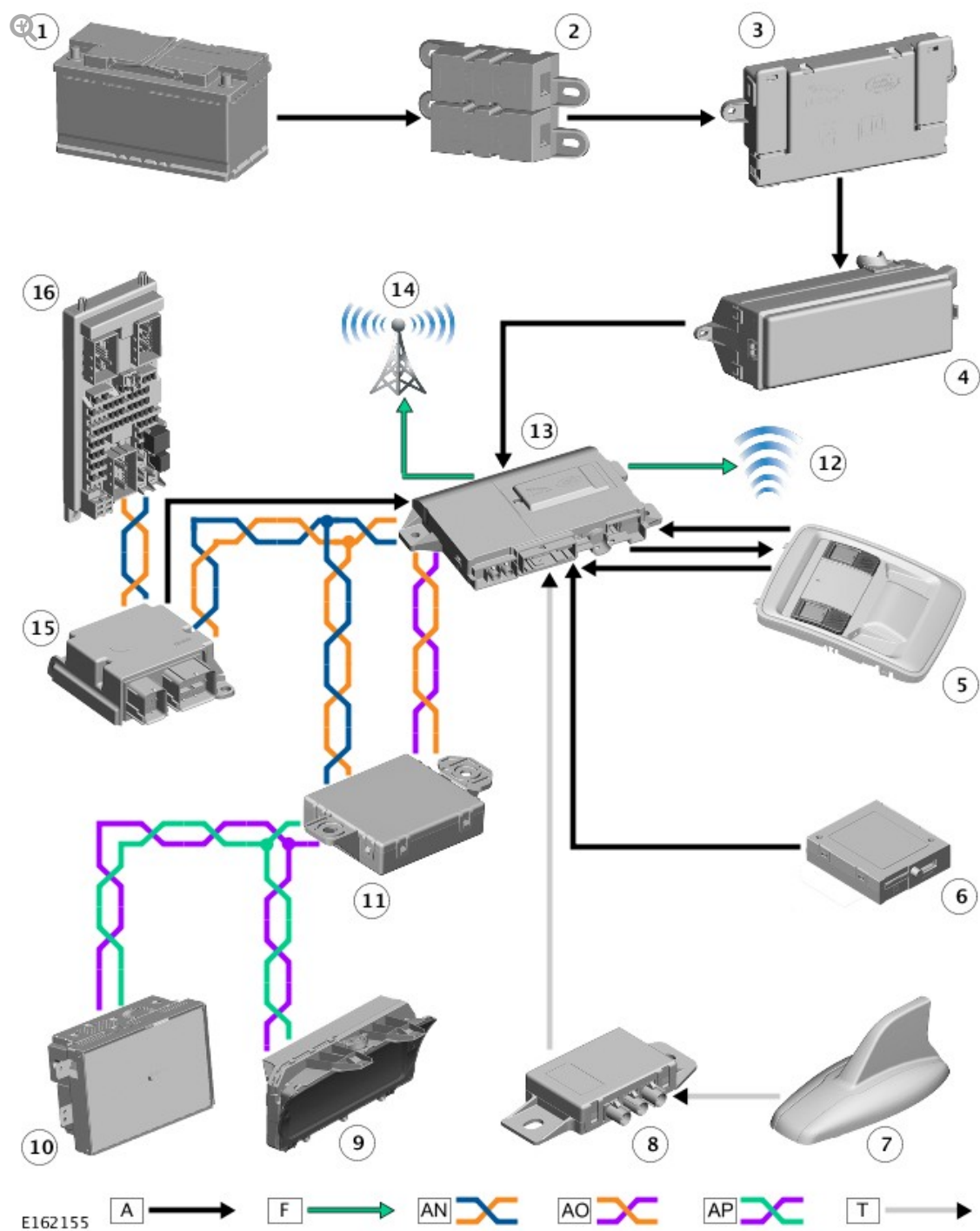
Check TCU and SIM Card Module for DTC's if error not in above list.

Use approved Land Rover diagnostic equipment to check for DTC's.

CONTROL DIAGRAM

NOTE:

A = Hardwired; **F** = Radio Frequency (RF) transmission (WiFi); **AN** = High Speed (HS) Controller Area Network (CAN) powertrain systems bus; **AO** = Medium Speed (MS) CAN body systems; **AP** = MS CAN comfort systems; **T** = Co-axial



ITEM	DESCRIPTION
1	Primary battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Rear Junction Box (RJB)
5	Front overhead console
6	SIM Card Module

7	Roof pod
8	GPS signal splitter
9	Instrument Cluster (IC)
10	Touch Screen (TS)
11	GateWay Module (GWM)
12	Wi-Fi - RF signal - internal antenna
13	Telematics Control Module (TCU)
14	GSM back-up signal - internal antenna
15	Restraints Control Module (RCM)
16	Central Junction Box (CJB)